

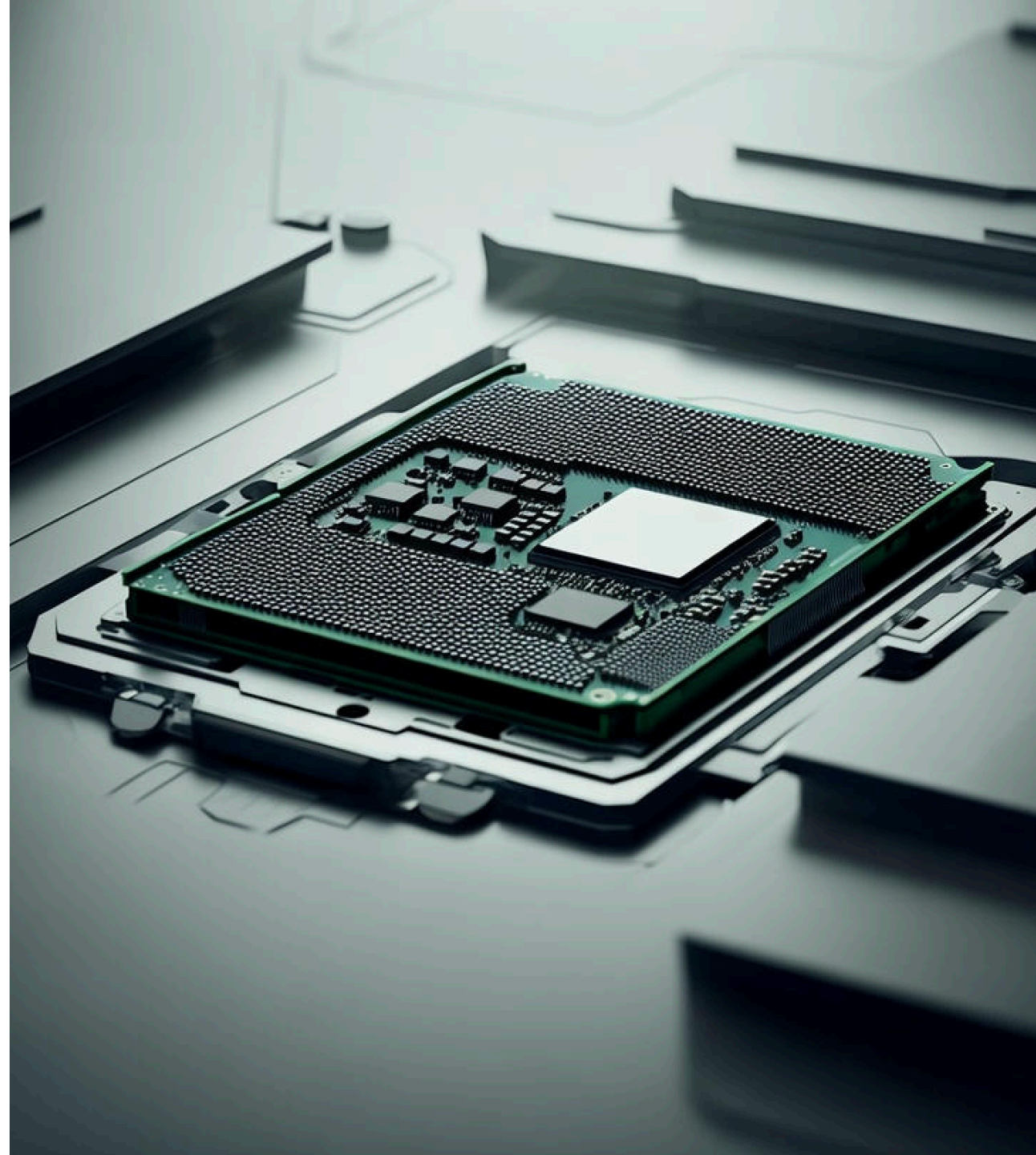
# SIMD-accelerated data processing

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blog: <https://lemire.me>

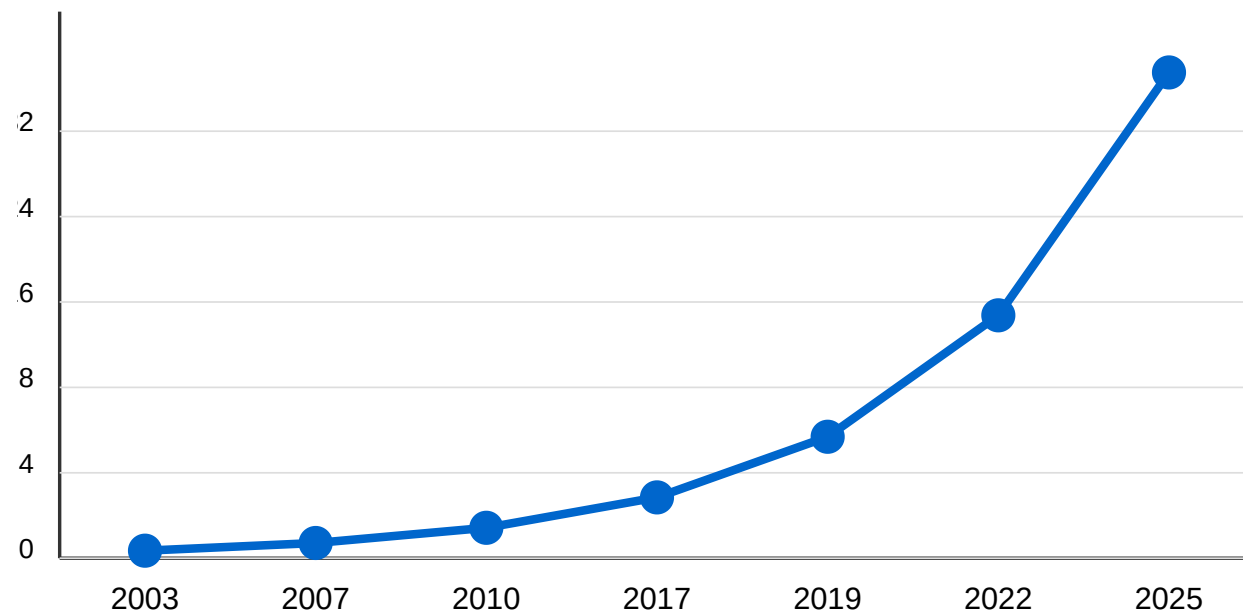
X: [@lemire](#)

GitHub: <https://github.com/lemire/>



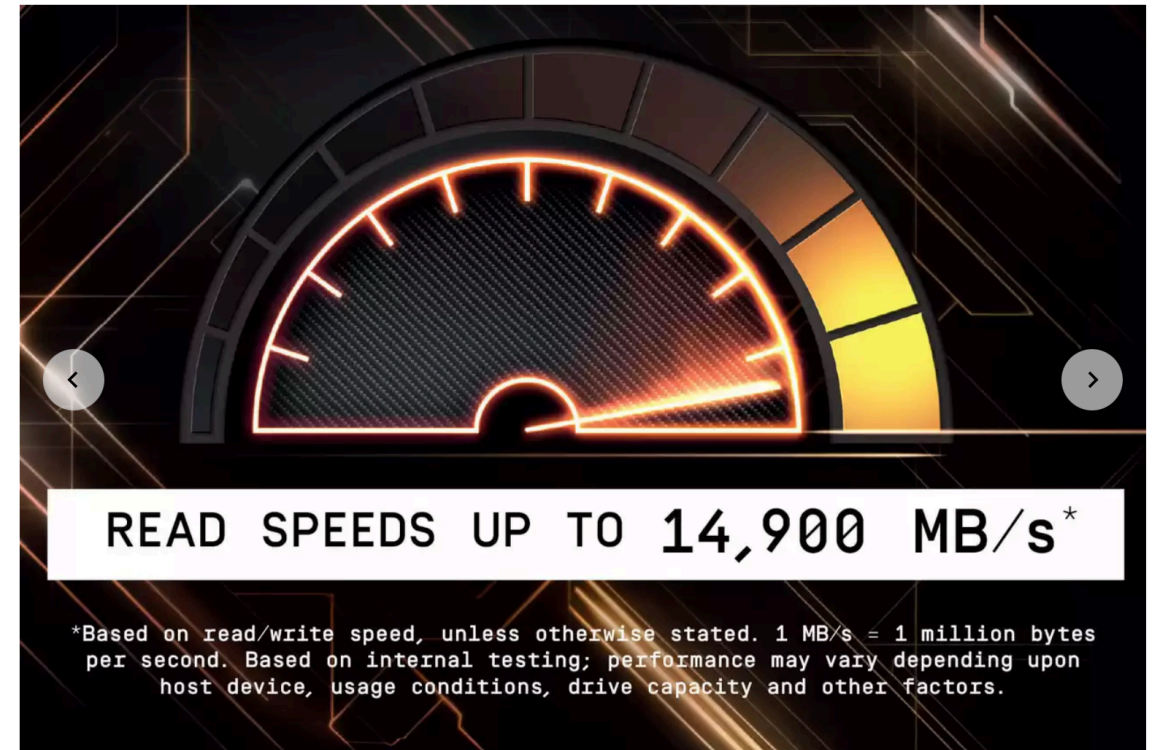
# Doubles every 3 years

| Specification | year | one channel |
|---------------|------|-------------|
| PCIe 1.x      | 2003 | 500 MB/s    |
| PCIe 2.x      | 2007 | 1 GB/s      |
| PCIe 3.x      | 2010 | 2 GB/s      |
| PCIe 4.x      | 2017 | 4 GB/s      |
| PCIe 5.x      | 2019 | 8 GB/s      |
| PCIe 6.x      | 2022 | 16 GB/s     |
| PCIe 7.x      | 2025 | 32 GB/s     |



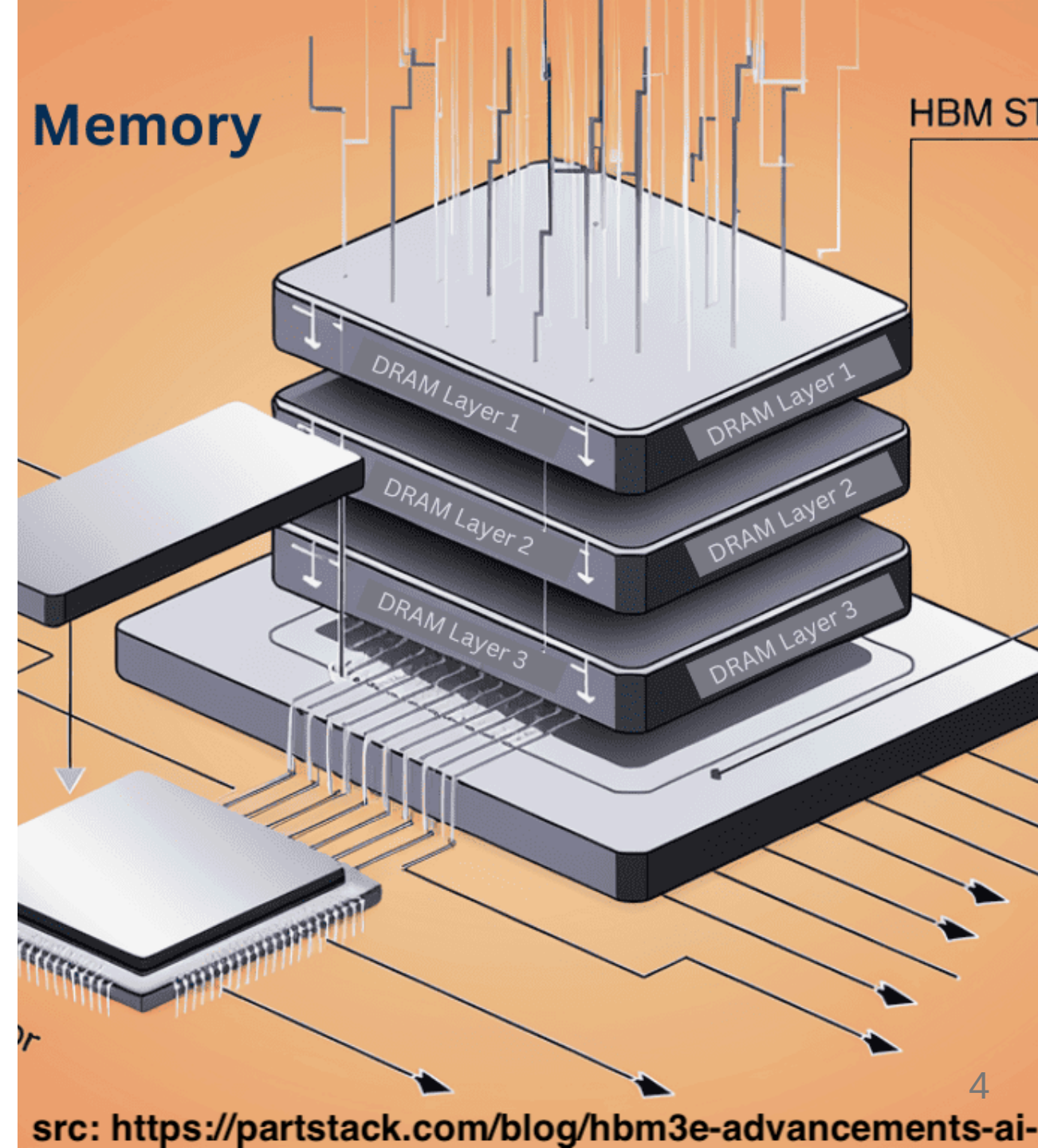
# Disk at gigabytes per second

- Sony PlayStation 5 (2020):  
5 GB/s
- Sony PlayStation 6 (2027):  
15 GB/s (?)



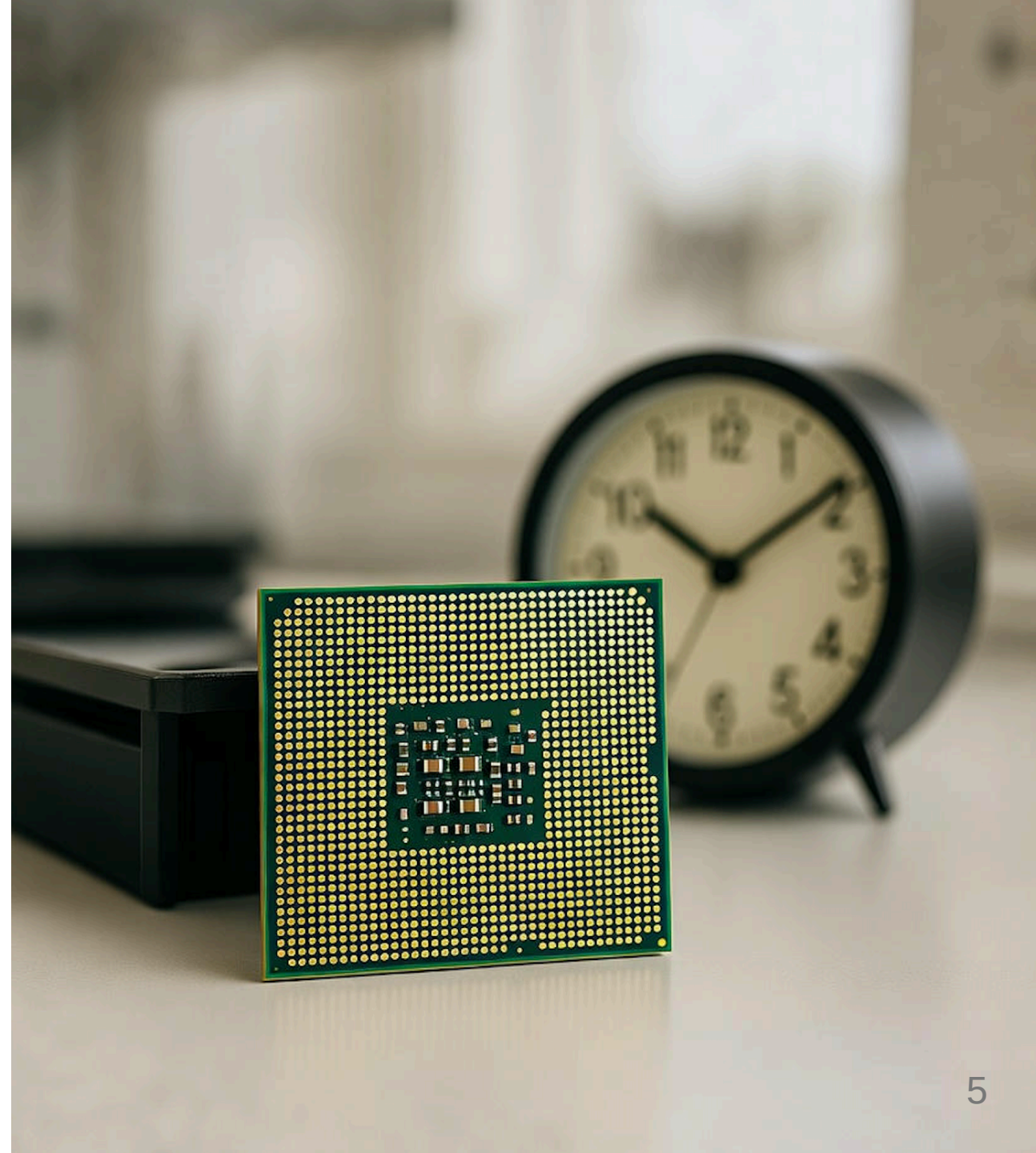
# High Bandwidth Memory

- Xeon Max processors contain 64 GB of HBM
- Bandwidth 800 GB/s



# Some numbers

- Processors: 4 GHz
- One byte per cycle: 4 GB/s
- **Easily CPU bound**



# JSON

- Portable, simple
- Used by ~97% of API requests.  
[Landscape of API Traffic 2021 - Cloudflare](#)
- scalar values
  - strings (must be escaped)
  - numbers (but not `NaN` or `Inf` )
- composed values
  - objects (key/value)
  - arrays (list)

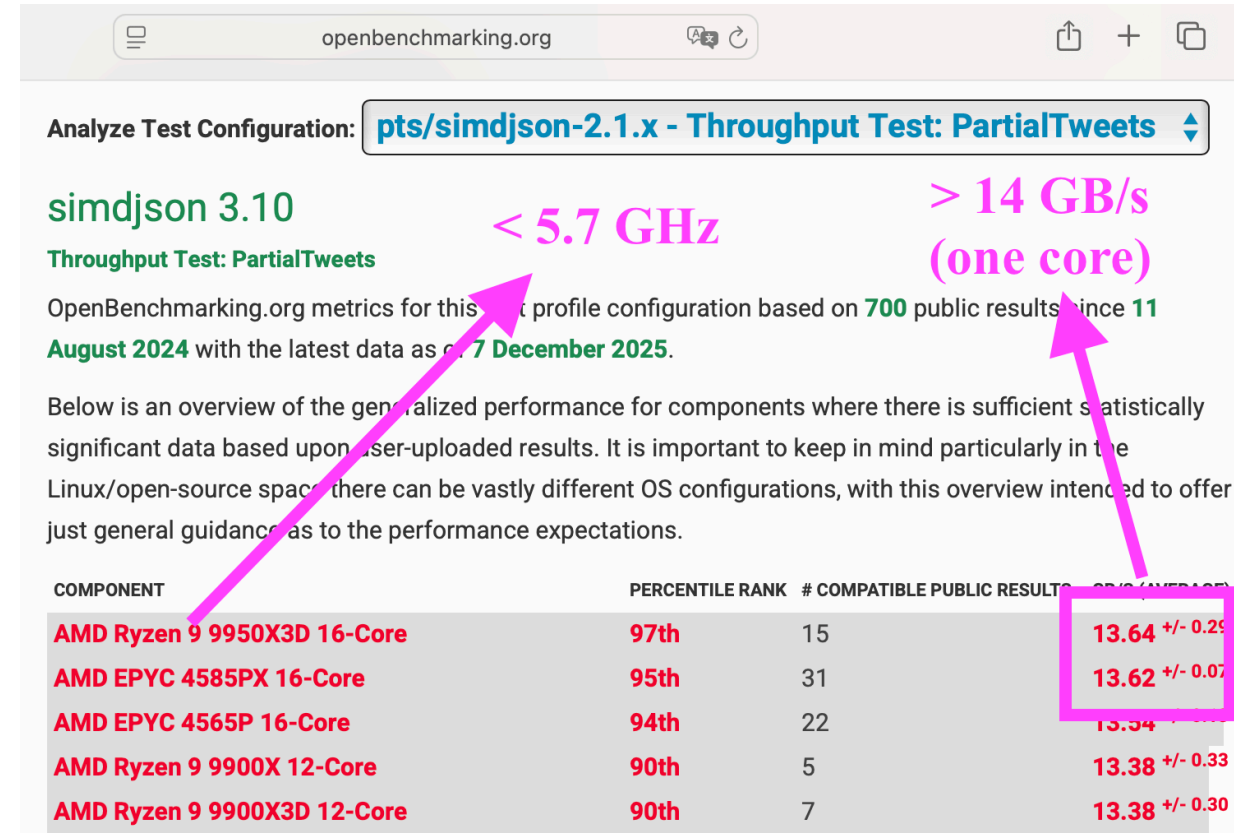
# JSON Downside?

Reading and writing JSON can be *slow*. E.g., 100 MB/s to 300 MB/s.

- Slower than fast disks or fast networks

```
$ go run parse_twitter.go  
Parsed 0.63 GB in 6.961 seconds (90.72 MB/s)
```

- openbenchmarking.org
- 14 GB/s, less than 5.7 GHz
- Parsing JSON at better than 2.5 bytes per cycle



## **SIMD (Single Instruction, multiple data)**

- Allows us to process 16 (or more) bytes or more with one instruction
- Supported on all modern CPUs (phone, laptop)
- Data-parallel types (SIMD) (recently added to C++26)

| <b>processor</b> | <b>year</b> | <b>arithmetic logic units</b> | <b>SIMD units</b> |
|------------------|-------------|-------------------------------|-------------------|
| Pentium 4        | 2000        | 2                             | $2 \times 128$    |
| AMD Zen 2        | 2019        | 4                             | $2 \times 256$    |
| AMD Zen 5        | 2024        | 6                             | $4 \times 512$    |

# SIMD Support in simdjson

- x64: SSSE3 (128-bit), AVX-2 (256-bit), AVX-512 (512-bit)
- ARM NEON
- POWER (PPC64)
- Loongson: LSX (128-bit) and LASX (256-bit)
- RISC-V: *upcoming*



# You are probably using simdjson

- Node.js, Electron,...
- ClickHouse
- WatermelonDB, Apache Doris, Meta Velox, Milvus, QuestDB, StarRocks

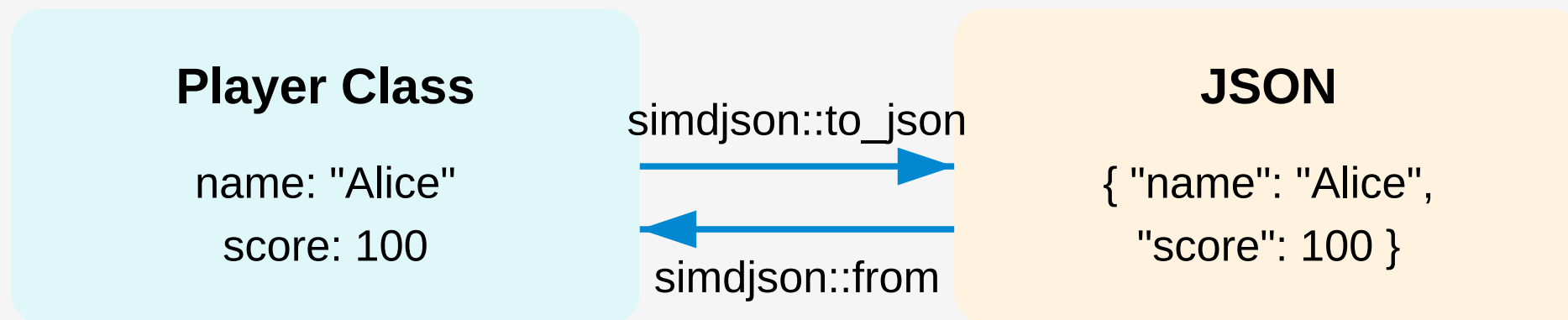


```
{
  "nom": "Marie Dubois",
  "age": 28,
  "ville": "Paris",
  "loisirs": ["lecture", "voyage", "yoga"],
  "contact": {
    "email": "marie.dubois@example.com",
    "phone": "+33123456789"
  }
}
```

# simdjson: Design

- First scan identifies the structural characters, start of all strings at about 10 GB/s using SIMD instructions.
- Validates Unicode at 30 GB/s.
- Rest of parsing relies on the generated index.
- Allows fast skipping. (Only parse what we need)
- Can minify JSON at 10 to 20 GB/s

# C++26 (compile-time reflection)



# Automatic Deserialization (C++26)

```
struct Player {  
    \\ ....  
}  
  
// Deserialization - one line!  
Player load_player(std::string& json_str) {  
    return simdjson::from(json_str); // That's it!  
}
```

# Automatic Serialization (C++26)

```
// Serialization - one line!  
void save_player(const Player& p) {  
    std::string json = simdjson::to_json(p); // That's it!  
    // Save json to file...  
}
```

# Classifying characters

- comma (0x2c) ,
- colon (0x3a) :
- brackets (0x5b,0x5d, 0x7b, 0x7d): [, ], {, }
- white-space (0x09, 0x0a, 0x0d, 0x20)
- others

# Vectorized classification

- Most SIMD ISAs support 'vectorized lookup tables' (at least 16-element)
- If we had 256-element tables, we could do `H(c)` .
- For 16-element tables, need two tables `H1` and `H2` .
- Find two tables `H1` and `H2` such as the bitwise AND of the look classify the characters: `H1(low(& 0xf) & H2(c >> 4)`

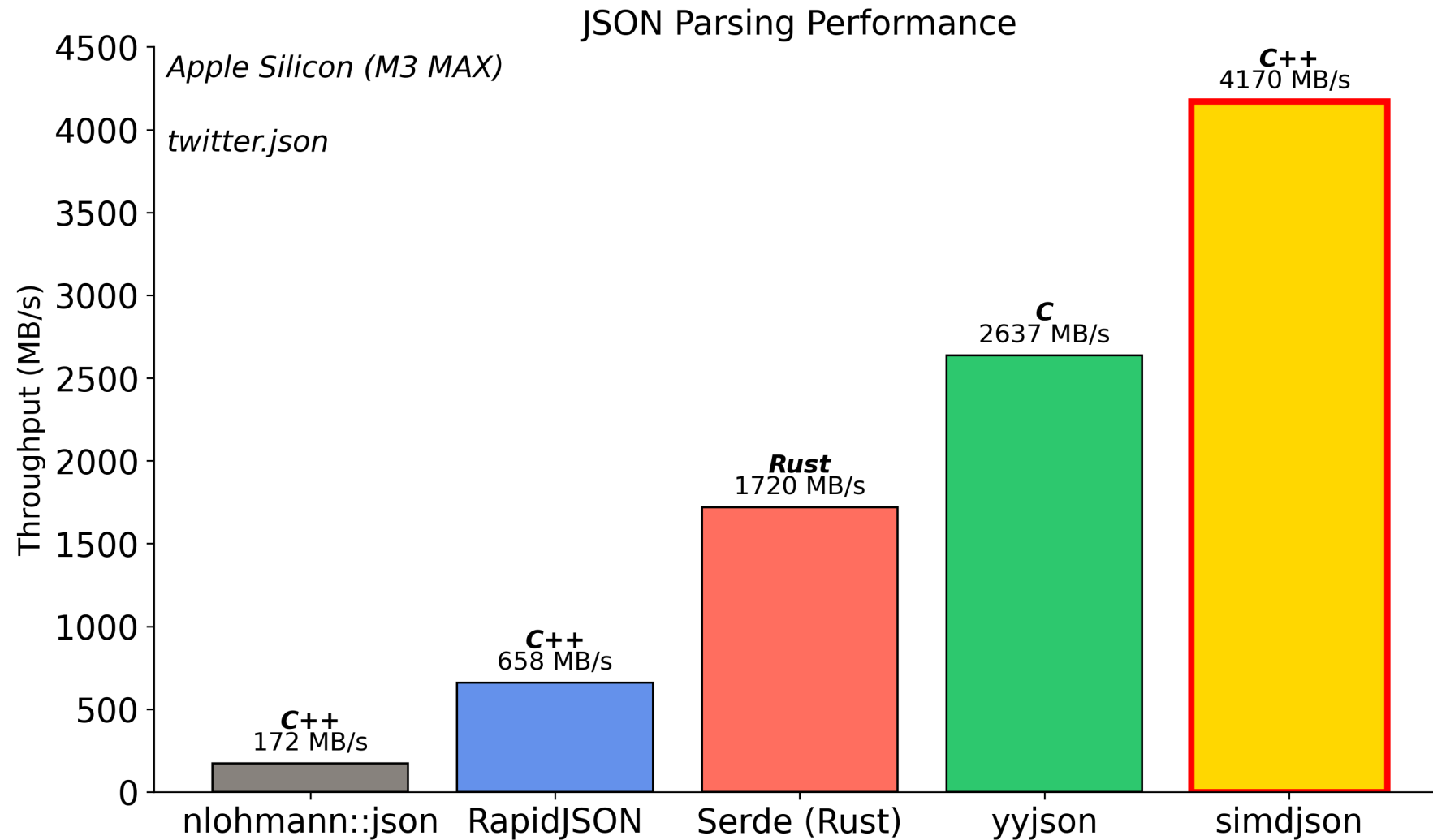
```
low_nibble_mask = {16, 0, 0, 0, 0, 0, 0, 0, 0, 8, 12, 1, 2, 9, 0, 0};  
high_nibble_mask = {8, 0, 18, 4, 0, 1, 0, 1, 0, 0, 0, 3, 2, 1, 0, 0};
```

Five instructions:

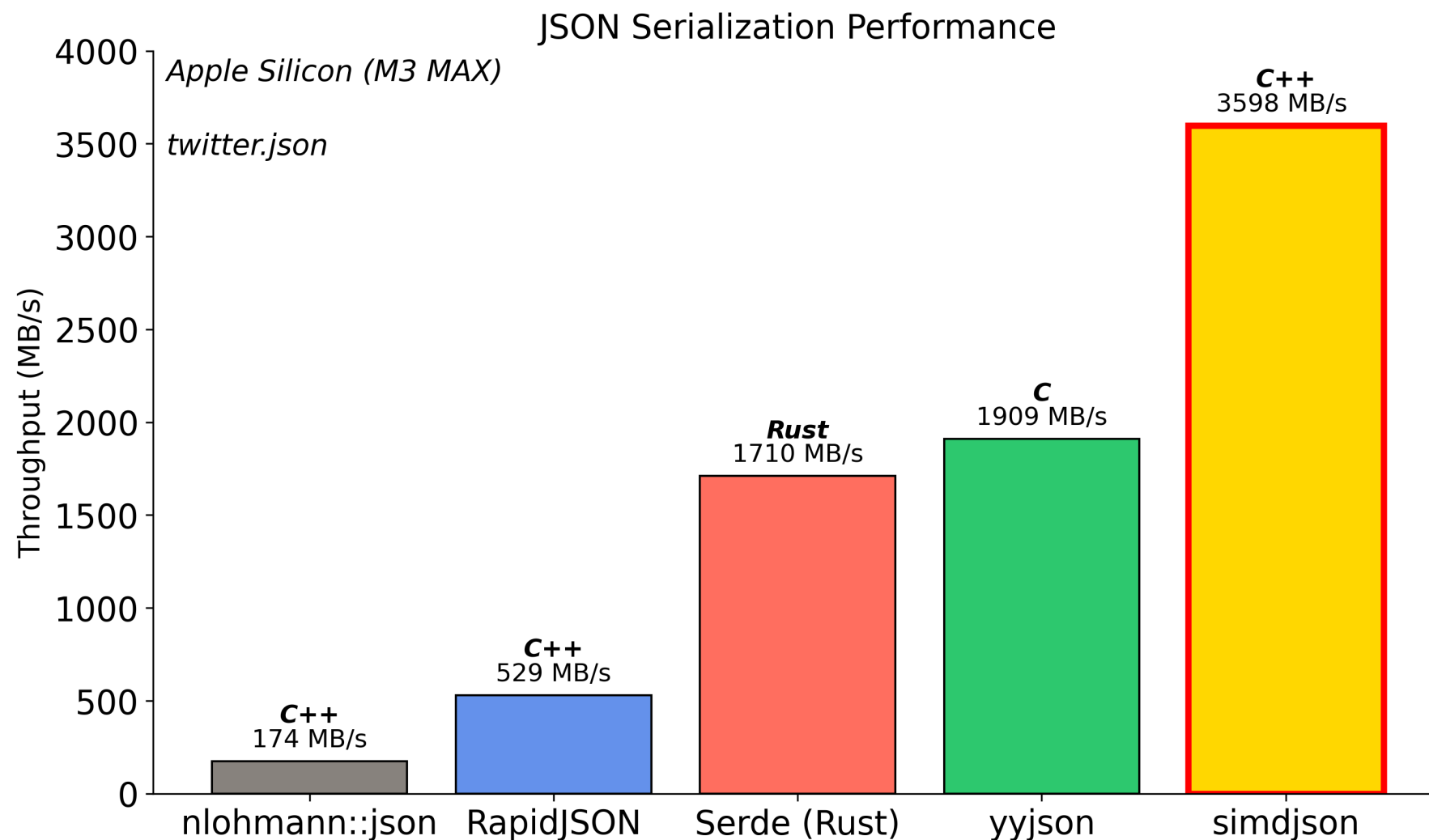
```
nib_lo = input & 0xf;  
nib_hi = input >> 4;  
shuf_lo = lookup(low_nibble_mask, nib_lo);  
shuf_hi = lookup(high_nibble_mask, nib_hi);  
return shuf_lo & shuf_hi;
```

- comma (0x2c): 1
- colon (0x3a): 2
- brackets (0x5b,0x5d, 0x7b, 0x7d): 4
- most white-space (0x09, 0x0a, 0x0d): 8
- white space (0x20): 16
- others: 0

# Deserialization (Apple Silicon)



# Serialization (Apple Silicon)



# Optimization #1: Consteval

## The Power of Compile-Time

**The Insight:** JSON field names are known at compile time!

**Traditional (Runtime):**

```
// Every serialization call:  
write_string("\"username\""); // Quote & escape at runtime  
write_string("\"level\"");    // Quote & escape again!
```

**With Consteval (Compile-Time):**

```
constexpr auto username_key = "\"username\""; // Pre-computed!  
b.append_literal(username_key); // Just memcpy!
```

# Optimization #2: SIMD String Escaping

**The Problem:** JSON requires escaping `"`, `\`, and control chars

**Traditional (1 byte at a time):**

```
for (char c : str) {  
    if (c == '"' || c == '\\ ' || c < 0x20)  
        return true;  
}
```

**SIMD (16 bytes at once):**

```
auto chunk = load_16_bytes(str);  
auto needs_escape = check_all_conditions_parallel(chunk);  
if (!needs_escape)  
    return false; // Fast path!
```

- Part of Safari, Chrome, and most browsers
- Process Unicode and Base64 formats at gigabytes per second
- Support LoongArch, x64, ARM, POWER, RISC-V



**simdutf**

# Unicode (UTF-16)

- Code points from U+0000 to U+FFFF, a single 16-bit value.
- Beyond: a surrogate pair [U+D800 to U+DBFF] followed by U+DC00 to U+DFFF

# Validate

- Check whether we have a lone code unit ( $x \leq 0xD7FF \vee x \geq 0xDBFF$ ), if so ok
- Check whether we have the first part of the surrogate ( $0xD800 \leq x \leq 0xDBFF$ ) and if so check that we have the second part of a surrogate

# Validate

```
PROCEDURE validate_utf16(code_units)
  i ← 0
  WHILE i < |code_units|
    unit ← code_units[i]
    IF unit ≤ 0xD7FF OR unit ≥ 0xE000 THEN
      INCREMENT i
      CONTINUE
    IF unit ≥ 0xD800 AND unit ≤ 0xDBFF THEN
      IF i + 1 ≥ |code_units| THEN
        RETURN false
      next_unit ← code_units[i + 1]
      IF next_unit < 0xDC00 OR next_unit > 0xDFFF THEN
        RETURN false
      i ← i + 2 // Valid surrogate pair
      CONTINUE
    RETURN false
  RETURN true
```

## toWellFormed()

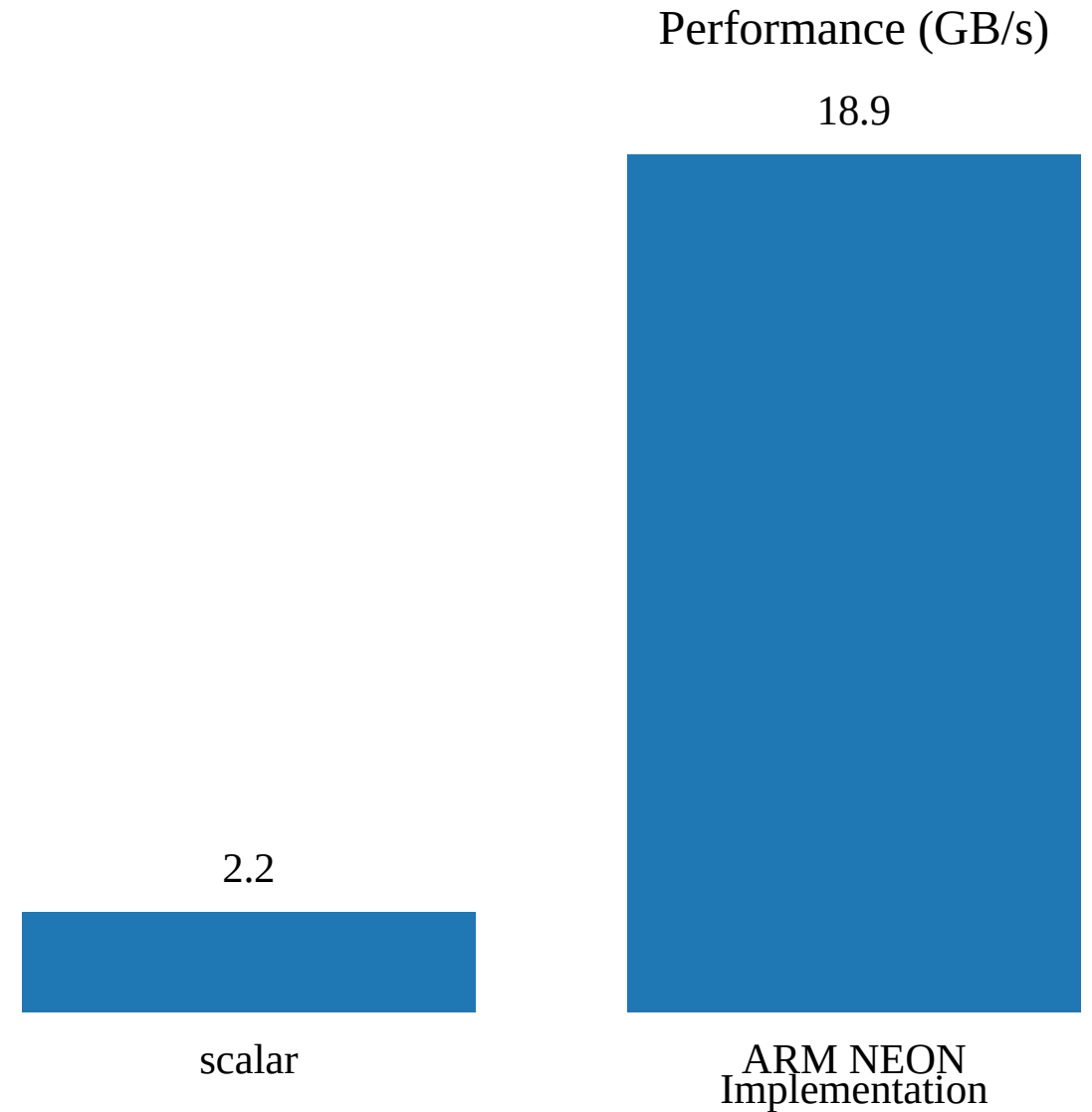
```
const str = "ab\uD800";  
console.log(str.toWellFormed());  
// "ab𐀀"
```

# UTF-16

- Write SIMD correction function (not just validation)
- Actually deployed in v8 (Google Chrome, Microsoft Edge)

## UTF-16 correction, Apple M4

|          | scalar | ARM NEON |
|----------|--------|----------|
| GB/s     | 2.2    | 18.9     |
| ins/byte | 12.0   | 0.9      |



## In Browser (Apple M4)

- Chromium : 16 GB/s (**uses our new function**)
- Firefox : 3.4 GB/s
- Safari : 1.2 GB/s

<https://lemire.github.io/browserwellformed/>

# Base64

- Encodes binary data to text using 64 characters (A-Z, a-z, 0-9, +, /)
- 3 bytes input → 4 characters output (33% overhead)
- Used in data URLs, email, web APIs

# Example

- `text = "Hello, World!"`

```
SGVsbG8sIFdvcmxkIQ==
```

# New JavaScript functions

```
const b64 = Uint8Array.prototype.toBase64(bytes); // string
const recovered = Uint8Array.prototype.fromBase64(b64); // Uint8Array, matches original 'bytes'
```

- SIMD accelerates encoding/decoding to gigabytes per second
- Part of simdutf: fast, portable implementations

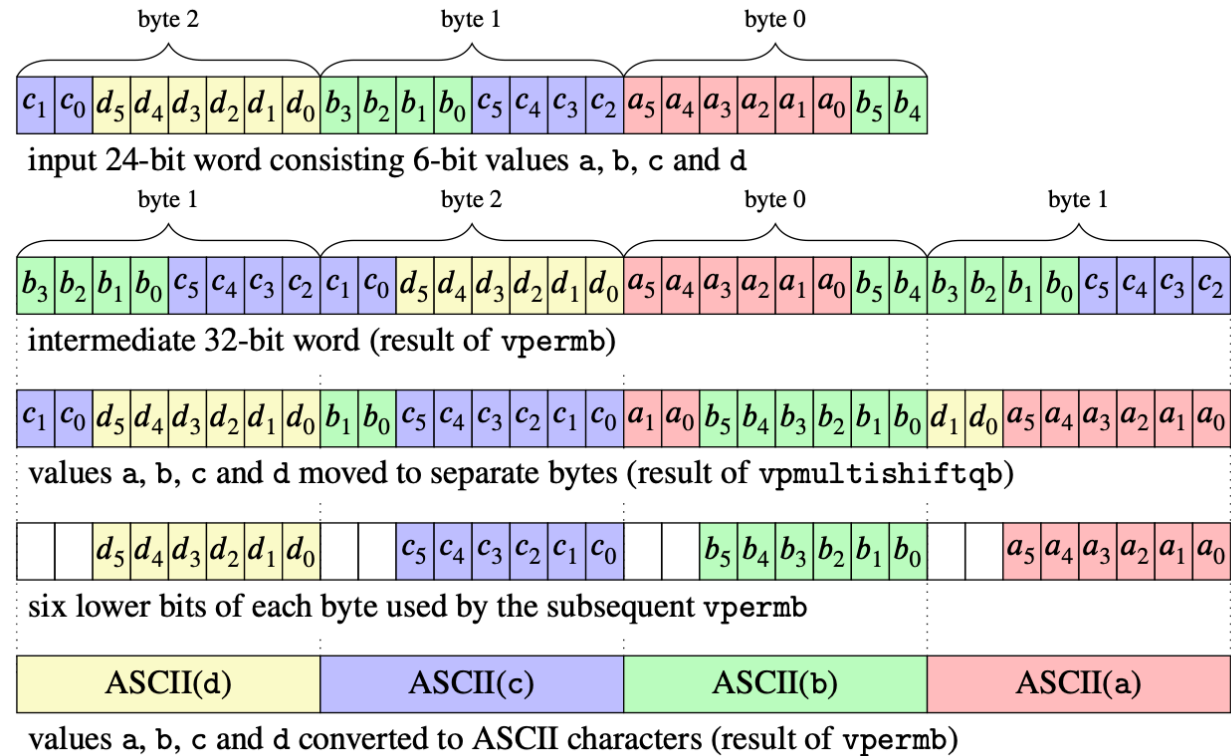
## Result in the browser (Safari, Apple M4)

| function                              | speed    |
|---------------------------------------|----------|
| <code>Uint8Array.fromBase64( )</code> | 11 GiB/s |
| <code>Uint8Array.toBase64( )</code>   | 20 GiB/s |

Test in your browser at <https://simdutf.github.io/browserbase64/>

# AVX-512 base64 encoding/decoding

- Encoding a 64-byte block requires only two non-memory instructions `vpermb` (twice) and `vpmultishiftqb`.



# Interested? Check these projects

- simdjson: The fastest JSON parser in the world <https://simdjson.org>
  - Node.js, Electron,...
  - ClickHouse, WatermelonDB, Apache Doris, Meta Velox, Milvus, QuestDB, StarRocks
- simdutf: Unicode routines (UTF8, UTF16, UTF32) and Base64 <https://github.com/simdutf/simdutf>
  - Node.js, Bun, WebKit (Safari), Chromium (Chrome, Edge)

# Credit

- simdjson reflection work with Francisco Geiman Thiesen (Microsoft)
- simdutf UTF-16 correction is joint work with Robert Clausecker
- simdjson and simdutf are community efforts (Geoff Langdale, John Keiser, Paul Dreik, Yagiz Nizipli and others)





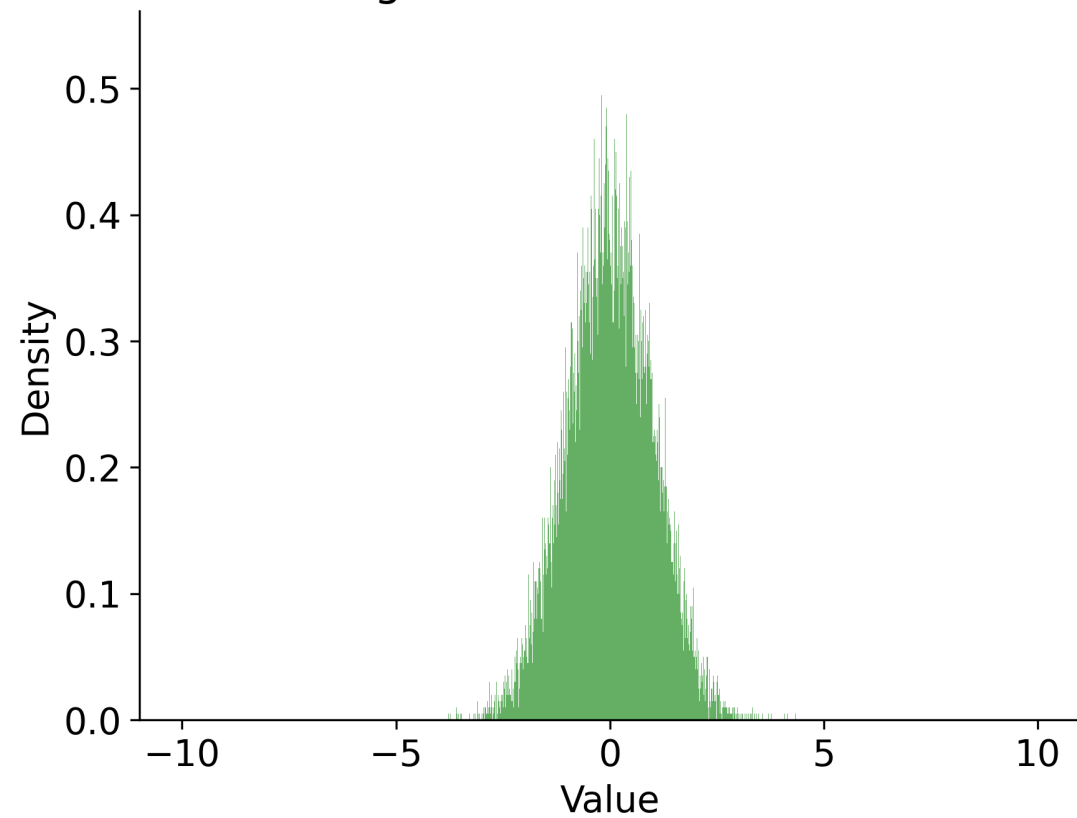
# Measurements

- We often assume that measurements (timings) are normally distributed.
- It is often an incorrect assumption.

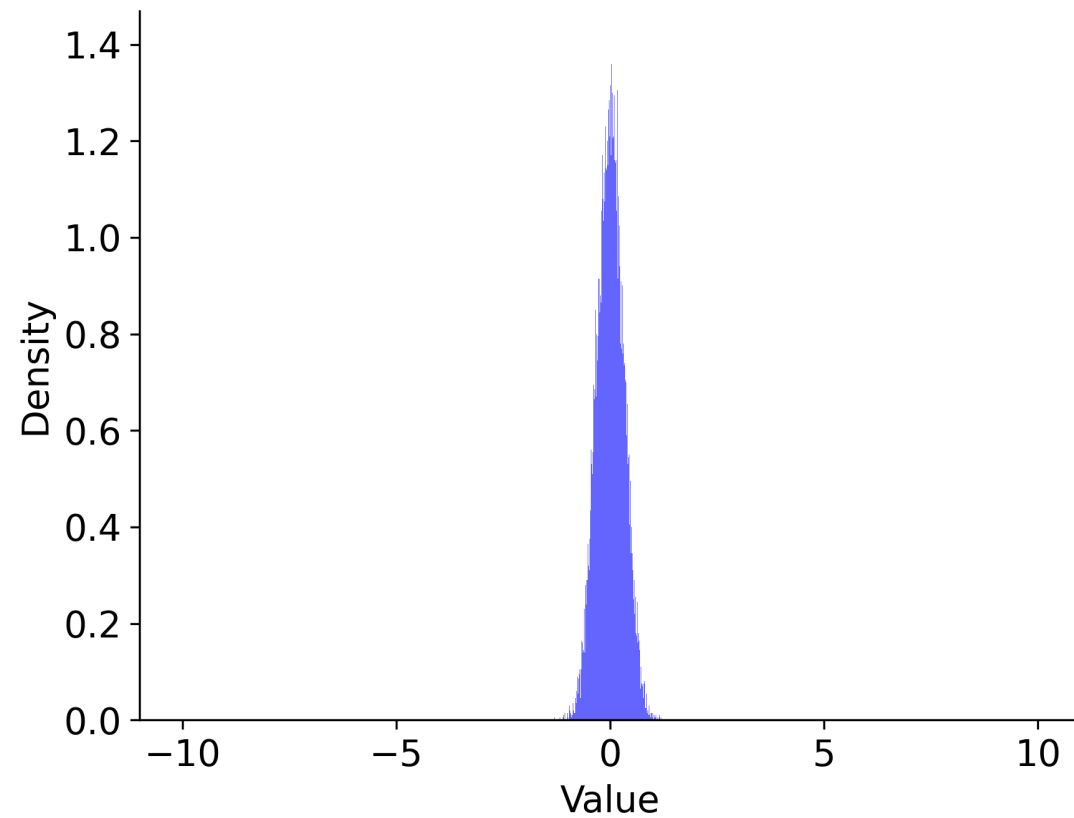
# Measurements

- If your measurements are normally distributed, the 'error' falls off as  $1/\sqrt{N}$

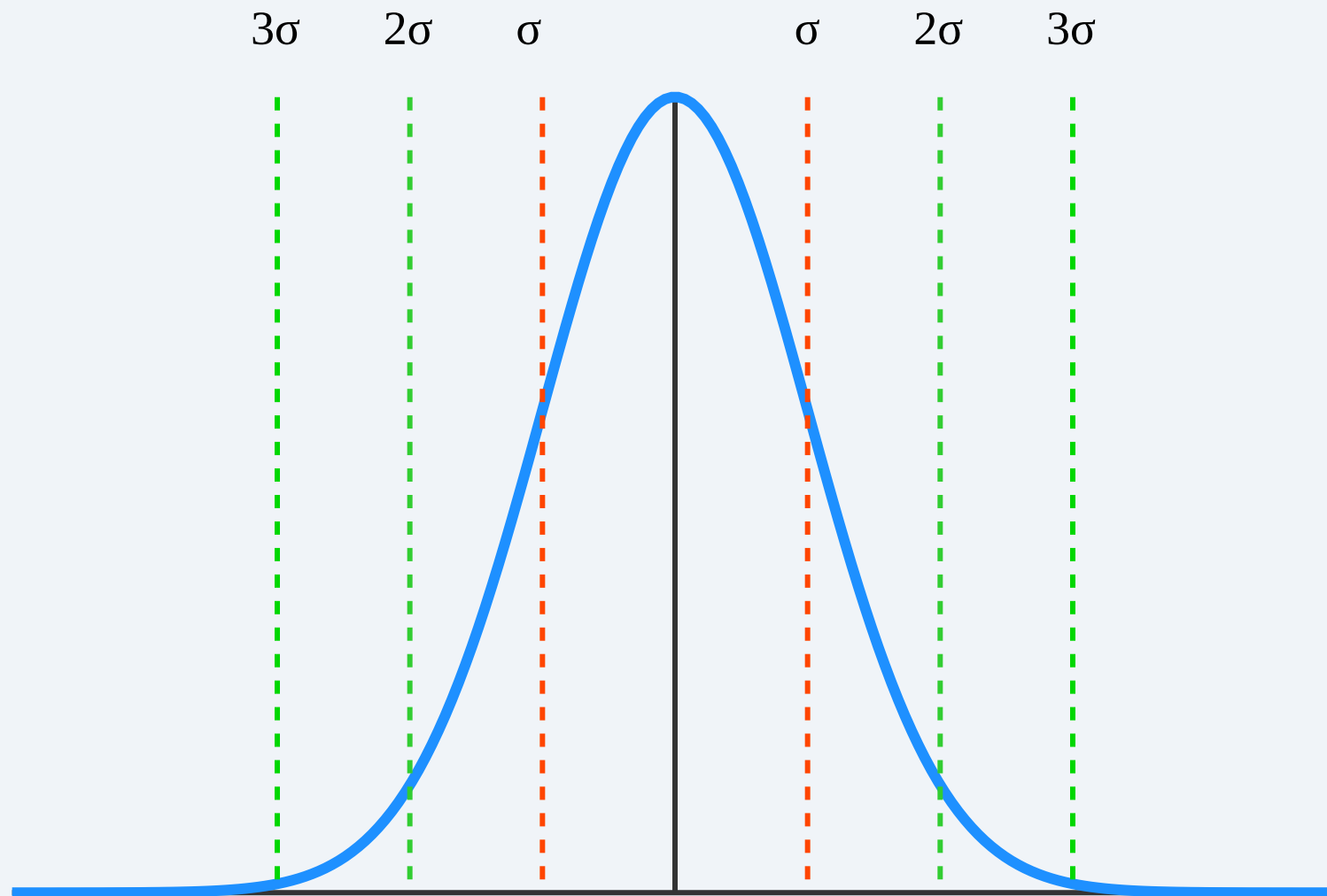
Original normal distribution



Sum of 10 i.i.d. normals / 10

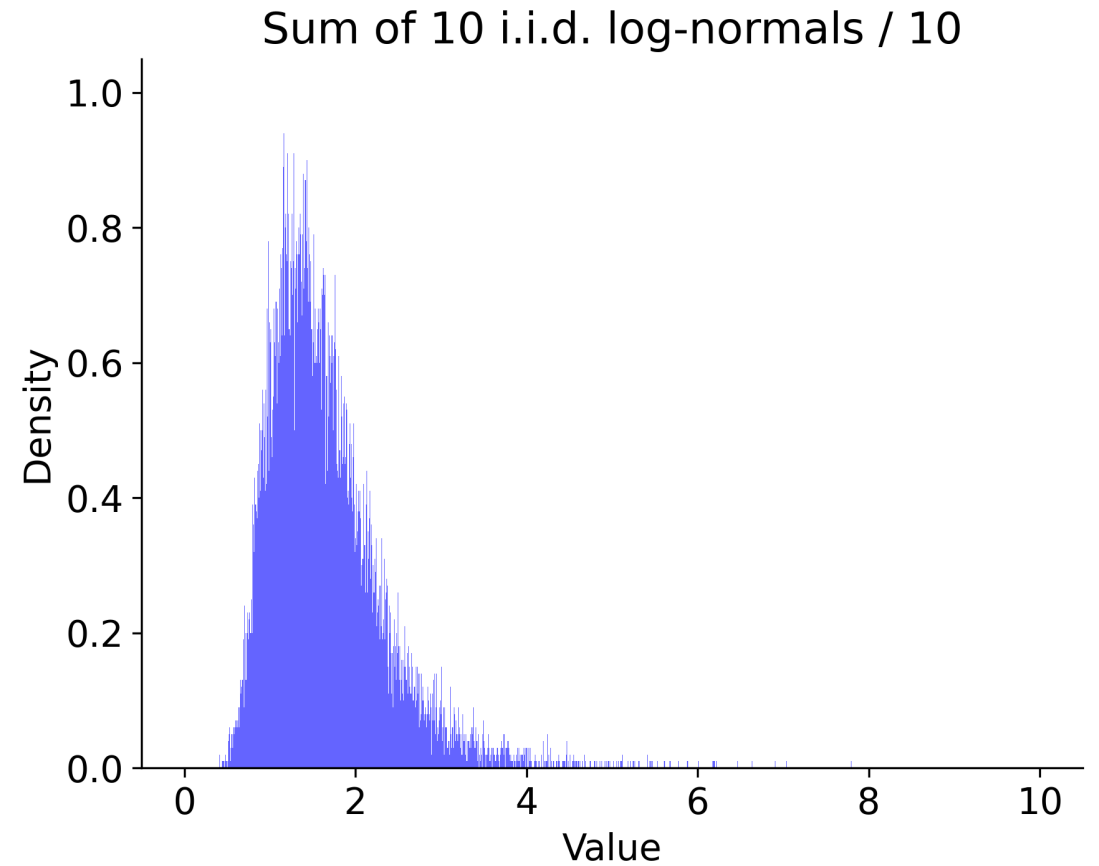
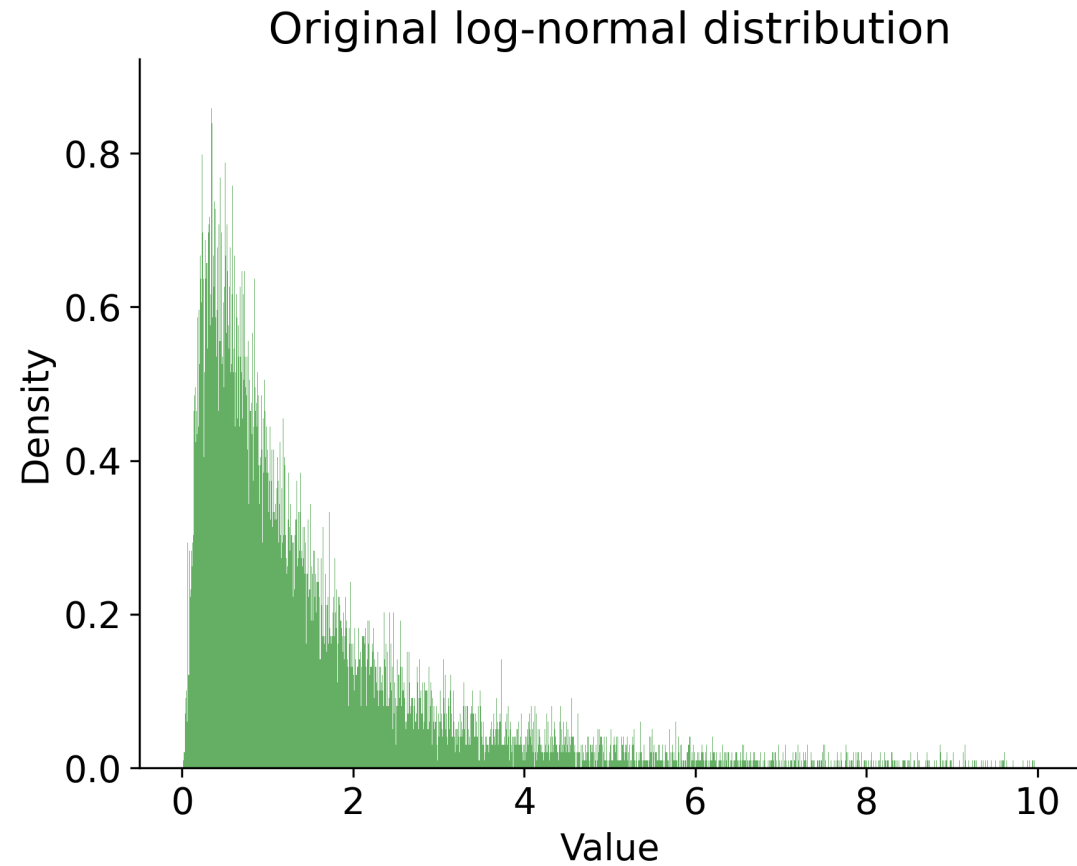


# Sigma events



- 1-sigma is 32%
- 2-sigma is 5%
- 3-sigma is 0.3% (once every 300 trials)
- 4-sigma is 0.00669% (once every 15000 trials)
- 5-sigma is 5.9e-05% (once every 1,700,000 trials)
- 6-sigma is 2e-07% (once every 500,000,000)
- $e^{-n^2/2} / (n * \sqrt{\pi/2}) \times 100$  for  $n > 3$

# What if we dealt with log-normal distributions?



# Real-world measurements

- You cannot assume normality
- Measurements are **not independent**.
- Reality: the absolute minimum is often a *reliable metric*
- Margin: difference between mean and minimum

# Conclusion

- Processors are getting much better! Wider!
- 'hot spot' engineering can fail, better to reduce overall instruction count.
- Branchy code can do well in synthetic benchmarks, but be careful.